MEASURED HARDWARE RESULTS

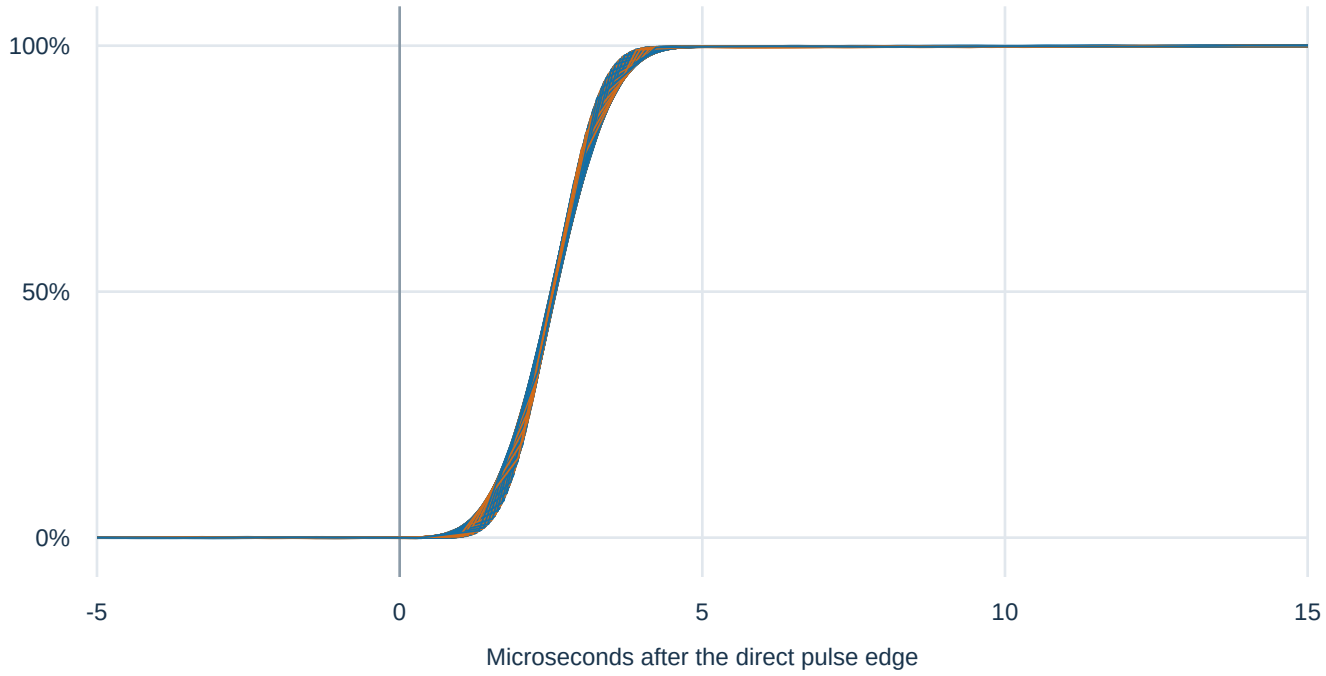
ADC pulse-delay measurement

Five completed trials | 354 transitions | 997 Hz square-wave input

Measured response delay: approximately 2.555 microseconds.
The measurement includes the analog input circuit, ADC response and FSYNC voltage translator, relative to the direct pulse at P4 GPIO6.

Transition	Count	Median delay (µs)	Std dev (ns)
Rising	176	2.5563	9.1
Falling	178	2.5547	7.6

Individual ADC step responses



Blue: rising **Orange: falling, inverted for comparison**
Each line joins individual ADC samples. Zero time is the direct hardware timestamp. All 354 traces are shown; no waveform-bin averaging is used.

Interpretation: the close agreement shows repeatability. The 7.6-9.1 ns scatter is **not an absolute accuracy bound**. P4 input-to-input timing calibration had not been performed when these results were collected.

METHOD AND TRACEABILITY

Trial details and measurement limits

Response-center medians below use the same individual readings as the graph.

Trial	Edges	Rising (µs)	Falling (µs)	FSYNC residual (ns)
1	71	2.5526	2.5525	4.74
2	69	2.5595	2.5578	5.49
3	70	2.5555	2.5540	4.48
4	72	2.5592	2.5554	5.28
5	72	2.5556	2.5537	5.19

Observed range across individual transitions
Rising: 2.5333-2.5797 µs. Falling: 2.5360-2.5795 µs.
FSYNC residual is deviation from fitted sample spacing, not absolute timing error.

Setup
ADS127L18 at 1,024,000 samples/s; 2.5 V reference; sinc4 filter, OSR 16. Rigol square wave: 997 Hz, 50% duty. Two equal **220-ohm resistors** reduced the CH0 branch. Operator-reported loaded levels: CH0P 0.128-1.46 V; generator/direct branch approximately -0.011-2.9 V. These scope readings are setup observations, not a simultaneous saved scope trace.

Timing method
GPIO6 captured the direct pulse; GPIO1 captured FSYNC after its voltage translator. Both used the same 160-MHz timer (6.25 ns per tick). The initial ADC frame was deliberately discarded; hardware pulse 2 identifies stored word 0. Each response used 40 individual readings. The response center is the average time of successive voltage changes, weighted by their signed sizes. Rising and falling results are reported separately.

What remains uncalibrated
Input threshold and capture-path differences between GPIO1 and GPIO6; divider and wiring effects; interpolation between approximately 976.6-ns-spaced ADC readings. FSYNC association relies on the checked startup assumption: spacing alone cannot exclude a constant whole-sample numbering error. No correction for these uncertainties was applied. This is the measured full-path response delay, not a chip-only filter-delay specification.

Source capture
captures/pulse-delay/20261105-153442/
result.json, p4.txt, feather.txt and comparison.html. All five trials completed.
SHA-256 of result.json: ff6d4ffd5c99ffd84138a0960a04652bad4adf9d34faf45c34bf3e8ea2d42200